



RAJESH VASHIST
CHIEF EXECUTIVE
OFFICER, SiTime

Microelectromechanical Systems

“SILICON Has REPLACED MANY ELEMENTS Being Used TRADITIONALLY”

The advent of nanotechnology and the likes is enabling lighter, smaller and cheaper electronics. This has led to the creation of microelectromechanical systems (MEMS) oscillators. Nidhi Arora, executive editor, *Electronics For You*, in conversation with Rajesh Vashist, chief executive officer, SiTime, discusses the technology behind creating MEMS, the benefits these bring and their adaptability in the Indian market

Q. What new technology features are being seen in MEMS oscillators being launched in the market?

A. Silicon-based MEMS oscillators work as an alternative to traditional quartz crystals and oscillators for clock timing. This is because MEMS oscillators provide the best dynamic performance by offering the best frequency stability. These can successfully maintain performance over temperature variation, harsh environments, shocks, vibrations and high power supply noise.

Silicon-based MEMS oscillators are sure to play a significant role in creating the next wave of technology. Since clock timing is a crucial element of any circuit, these will enable precision timing and continuous connections in tough environments. 5G phone services will have stringent requirements, and silicon-based MEMS are being developed to work with these. These deliver advanced driver assistance systems with uninterrupted satellite locking, low-power Internet of Things (IoT) for longer connectivity in a tiny package and integration into systems on chip (SoC) for high-volume production and ease of design.

Q. What are these features important for, and why?

A. Silicon has replaced many elements being used traditionally, making the end product better and highly efficient. These features are significant for innovators and makers

of new-age electronic products for both consumers and industries. These may include clock synchronisation in telecom, broadband, video and instrumentation, mobilephones, tablets, data cards, wearables, industrial solutions like high-performance equipment, small cells, industrial GPS, satellites, IEEE 1588 (clock synchronisation protocol for systems on a network) and synchronous optical networking.

Silicon-based MEMS can help reduce electromagnetic interference in systems, whether industrial, office automation or consumer. Voltage-controlled crystal oscillators can be replaced by SiTime's lower noise digitally-controlled oscillators.

Q. Are there any challenges associated with silicon-based MEMS oscillators?

A. Silicon-based oscillators are relatively new as compared to crystals that have been in use since decades. Although available for most basic technologies, research is in progress to increase the frequency range for these. Wider ranges are yet to be launched, and the product base is still expanding.

Also, radiation-based hardened range is yet to be developed based on research analysis. This will be highly useful in space-related applications.

Q. What is the impact on pricing with respect to the available range?

A. Silicon-based MEMS oscilla-

tors are competitively priced with crystal oscillators available in the market. As demand for these further increases and economies of scale are brought about, variation gap in high ranges will reduce. Higher frequencies and smaller sizes are generally higher priced.

Q. What is the road ahead for silicon-based MEMS oscillators?

A. With the unique benefits these oscillators bring—like smallest size, low power, high stability, dynamic performance and vibration immunity—for various applications in industrial, mobile, consumer, automotive and so on, along with 20 times better shock absorption, 30 times more reliability (1.2 billion hours mean time between failures), silicon-based MEMS oscillators are the future. No other technology is disrupting the timing industry.

Q. What are the unique benefits provided by SiTime's MEMS device range?

A. SiTime offers better quality and higher reliability than quartz resonators. Good shock and vibration resistance, support for non-standard frequency values and small size qualifies only one device that supports multiple frequencies.

For advanced driver assist systems, SiTime offers a small footprint of 2.0mm × 1.6mm, higher temperature range, better long-term aging and competitive price. **EFY**